

DatabaseFinalProject

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Designing an Efficient Recipe Management database

When creating this database, the primary goal was to design a well-structured and scalable database that efficiently manages recipes, ingredients, cooking hardware, and other related information. The database follows a normalized approach to avoid redundancy while ensuring flexibility in data storage and retrieval.

Key Design Decisions

1. Separation of Core Entities

The database is structured around the Recipe table, which holds the fundamental details of a recipe, such as its name, total cooking time, and category. Other related entities, like Ingredients, Instructions, Images, and Cooking Hardware, are linked to recipes to maintain clarity and efficiency.

2. Handling Many-to-Many Relationships

Several aspects of a recipe involve many-to-many relationships, such as ingredients and cooking hardware. To properly model this, junction tables like `Ingredients_In_Recipes` and `CookingHardware_In_Recipes` were introduced. This approach allows multiple ingredients and hardware items to be associated with multiple recipes without redundancy.

3. Alternative Ingredients and Recipes

One unique aspect of the design is the inclusion of alternative ingredients and recipes through the `Ingredients_And_Alternatives` and `AlternativeRecipes` tables. This ensures users can easily find substitutions for ingredients and explore variations of a recipe.

4. Allergy Warnings and Recipe Categorization

To enhance usability, an `AllergyWarnings` table is included to provide critical dietary information, and a `Category` table classifies recipes into different types (e.g., desserts, main courses, etc.), making it easier to filter and find relevant recipes.

SQL Queries and Data Population

Once the database structure was in place, we moved on to inserting data into the tables. This step was relatively straightforward but time-consuming, as we had to carefully consider which data would be relevant and useful, especially in the context of recipes. The ER diagram proved to be an essential guide, ensuring that all necessary information was included.

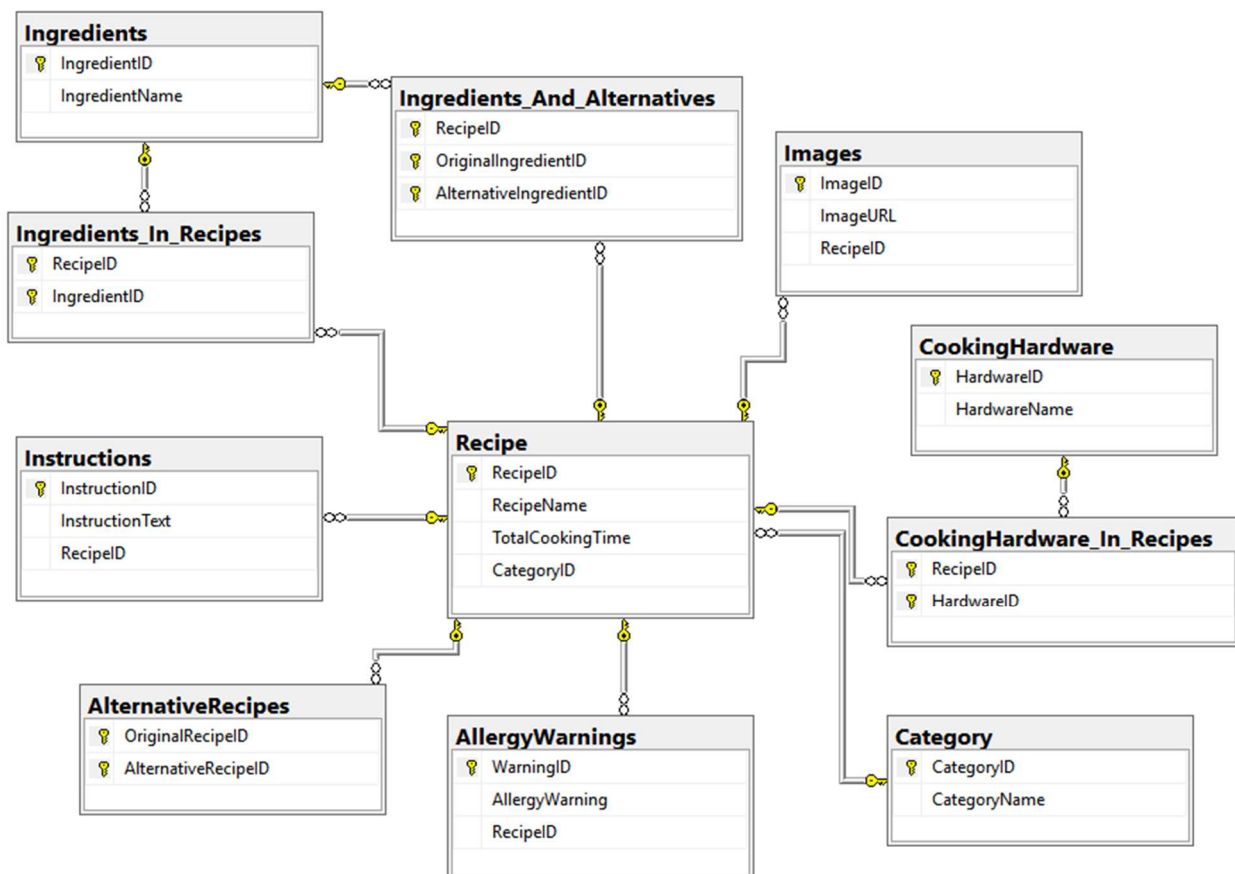
After completing the data insertion process, we faced another challenge: retrieving data efficiently using SQL queries. Constructing JOIN queries to fetch data from multiple tables was particularly tricky and required several hours of troubleshooting. However, through trial

and error, we managed to overcome these difficulties and successfully implemented functional queries to retrieve and manage data effectively.

Conclusion

This database was designed with efficiency, scalability, and user-friendliness in mind. By organizing data through proper normalization, avoiding redundancy, and incorporating useful features like ingredient alternatives and allergy warnings, the structure ensures a smooth and comprehensive experience for users managing or searching for recipes. The database lays a strong foundation for a robust recipe management system that can be expanded in the future.

ER Diagram



Example query's and results

```
-- Test retrieving all ingredients for a specific recipe (Spaghetti Bolognese)
SELECT i.IngredientName FROM Ingredients i
JOIN Ingredients_In_Recipes ir ON i.IngredientID = ir.IngredientID
WHERE ir.RecipeID = 3;
```

	IngredientName
1	Spaghetti
2	Ground Beef

```
-- Test retrieving all recipes under 'Dinner' category
```

```
SELECT r.RecipeName FROM Recipe r
JOIN Category c ON r.CategoryID = c.CategoryID
WHERE c.CategoryName = 'Dinner';
```

	RecipeName
1	Spaghetti Bolognese
2	Beef Stroganoff

```
-- Test retrieving all instructions for a recipe (Beef Stroganoff)
```

```
SELECT InstructionText FROM Instructions WHERE RecipeID = 8;
```

	InstructionText
1	Sauté onion in a skillet until soft. Add sliced beef sirloin and brown. Serve with rice or noodles.

```
-- Test retrieving allergy warnings for a recipe (Garlic Shrimp Appetizer)
```

```
SELECT AllergyWarning FROM AllergyWarnings WHERE RecipeID = 5;
```

	AllergyWarning
1	Contains shellfish